AP PRECALCULUS

Unit 1 • Worksheet 1.9-1.10A • Answer key

Worksheet 1.9-1.10A • Rational Functions and Asymptotes

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2 - x - 6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Solution

Answer: (B). $(x+2)$ cancels (hole at $x = -2$).

Question 2

NO CALCULATOR

Where is the hole in the graph of $r(x) = \dfrac{x^2 + x - 2}{x^2 - 1}$?

(A) $\left(1, \frac{3}{2}\right)$

(B) $(-1, 0)$

(C) $(1, 0)$

(D) $(-2, 0)$

Solution

Answer: (A). $\frac{(x+2)(x-1)}{(x-1)(x+1)}$: hole at $x = 1$, height $\frac{3}{2}$.

Question 3

NO CALCULATOR

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) ∞

(B) $-\infty$

(C) 2

(D) 0

Solution

Answer: (A). Numerator near 2, denominator positive and small.

Question 4

NO CALCULATOR

What happens on the graph of $r(x) = \dfrac{x-2}{(x-2)^2}$ at $x = 2$?

(A) a hole
(B) a zero
(C) a vertical asymptote
(D) nothing

Solution
Answer: (C). $r(x) = \frac{1}{x-2}$ for $x \neq 2$.

Question 5 (Free response)

NO CALCULATOR

For each function, find all vertical asymptotes and holes. For each asymptote, give both one-sided limits. (a) $r(x) = \dfrac{x-1}{(x+2)(x-3)}$ (b) $s(x) = \dfrac{x^3-4x}{x^2+2x}$ (c) $t(x) = \dfrac{x+1}{(x-2)^2}$

Solution
(a) Asymptotes $x = -2$: $\lim_{x\to -2^-} = -\infty$, $\lim_{x\to -2^+} = +\infty$; $x = 3$: $\lim_{x\to 3^-} = -\infty$, $\lim_{x\to 3^+} = +\infty$. No holes.
(b) $s = \frac{x(x-2)(x+2)}{x(x+2)} = x - 2$: holes $(0, -2)$ and $(-2, -4)$; no asymptotes.
(c) Asymptote $x = 2$; both one-sided limits $+\infty$. No holes.